Common notions and shared lemmas

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Common` contains the notions shared by all formalised results (Sections 1–2 and App. A of the paper) together with the auxiliary results that are used by more than one proof: Observation 21, Lemma 22, Lemma 29 and Eq. (3), the ℓ^p/operator norms with Lemma 19, the cut norm, the optimal transport distances and the attainment of their minima (App. A), the edit distance with Eq. (7), and the graphs of the reductions of Section 4. A reviewer-oriented overview of all definitions is `Common_overview.pdf`.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Sec. 2	5	Adjacency matrix $A_G \in \{0,1\}^{V\times V}$ (real entries)	`GsimOT.adj` `Common/Basic.lean:41`

App. A	14	Degree $\deg_G(v) = \lvert N_G(v)\rvert$; $\sum_{v'} A_G(v,v') = \deg_G(v)$	`GsimOT.deg`
			`Common/Basic.lean:45`
			`GsimOT.sum_adj_eq_deg`
			`Common/Basic.lean:69`

App. A, Eq. (6)	14	Discrepancy $D_{G,H}(v,w,v',w') = A_G(v,v') - A_H(w,w')$	`GsimOT.disc` `Common/Basic.lean:78`
Sec. 2	5	$\mathrm{DS}(V,W)$: doubly stochastic matrices	`GsimOT.IsDS` `Common/Basic.lean:85`

Sec. 2	5	Permutation matrix $P_\pi \in \mathrm{Perm}(V,W)$; $P_\pi \in \mathrm{DS}(V,W)$	`GsimOT.permMat`
			`Common/Basic.lean:91`
			`GsimOT.permMat_isDS`
			`Common/Basic.lean:93`

Sec. 2	5	Local cost $L_Q(v,w) = \sum_{(v',w')} \lvert D(v,w,v',w')\rvert\, Q(v',w')$	`GsimOT.localCost` `Common/Basic.lean:131`

Paper	Page	Statement	Lean declaration(s) and location

Sec. 2, Eq. (7)	5, 14	$C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$	`GsimOT.costOne`
			`Common/Basic.lean:137`
			`GsimOT.localCost_le_costOne`
			`Common/Basic.lean:176`
			`GsimOT.costOne_le`
			`Common/Basic.lean:182`

Sec. 2	5	$C_1(\pi) = C_1(P_\pi)$ and $L_{P_\pi}(v,\pi(v)) = \lvert\pi(N_G(v)) \triangle N_H(\pi(v))\rvert$	`GsimOT.costOneBij`
			`Common/Basic.lean:141`
			`GsimOT.localCost_permMat_apply`
			`Common/Basic.lean:207`
			`GsimOT.costOneBij_le_iff`
			`Common/Basic.lean:212`

Sec. 1, Sec. 2	1, 5	$d_1(G,H) = \min_{\pi\in\mathrm{Bij}(V,W)} C_1(\pi)$	`GsimOT.dOne` `Common/Basic.lean:144`
Sec. 2	5	$d_1^{\mathrm{OT}}(G,H) = \min_{Q\in\mathrm{DS}(V,W)} C_1(Q)$ (as an infimum over $\mathrm{DS}(V,W)$)	`GsimOT.dOTOne` `Common/Basic.lean:148`

Sec. 1, Sec. 2	1, 5	Normalisations $\delta_1 = n^{-1}d_1$, $\delta_1^{\mathrm{OT}} = n^{-1}d_1^{\mathrm{OT}}$ (scaling by $1/n$)	`GsimOT.deltaOne`
			`Common/Basic.lean:151`
			`GsimOT.deltaOTOne`
			`Common/Basic.lean:154`

Sec. 1, Sec. 3	3, 6	$d_1^{\mathrm{OT}} \le d_1$ (every bijection is a fractional alignment)	`GsimOT.dOTOne_le_dOne` `Common/Basic.lean:264`
Sec. 2, Eq. (3)	5	Birkhoff–von Neumann: $\mathrm{supp}(Q)$ contains the graph of a bijection	`GsimOT.exists_bij_subset_supp` `Common/Birkhoff.lean:20`

Observation 21	20	$L_Q(v,w) \ge \lvert\deg_G(v) - \deg_H(w)\rvert$	`GsimOT.claim26`
			`Common/Claim26_DegreeBound.lean:21`
			`GsimOT.claim26_bij`
			`Common/Claim26_DegreeBound.lean:40`

Paper	Page	Statement	Lean declaration(s) and location
Lemma 22 (construction)	21	Adding the clique K ($m = 2\Delta + 2\tau + k(\tau + \delta) + 1$) and the staircase	`GsimOT.lockSize` Common/Lemma27_LockingMarkers.lean:36 `GsimOT.stair` Common/Lemma27_LockingMarkers.lean:40 `GsimOT.lockGraph` Common/Lemma27_LockingMarkers.lean:56
Lemma 22 (1)	21	$C_1(\pi) < \tau \Rightarrow \pi$ fixes all markers and $\pi(V) = V$	`GsimOT.lemma27_bij` Common/Lemma27_LockingMarkers.lean:161
Lemma 22 (2)	21	$Q(z_j, z_j) = 1 \Rightarrow C_1(Q')$ w.r.t. (G', H') equals $C_1(Q)$ w.r.t. (G, H)	`GsimOT.extendId` Common/Lemma27_LockingMarkers.lean:260 `GsimOT.lemma27_cost` Common/Lemma27_LockingMarkers.lean:351
App. D.2	28	Pseudometric $\rho_H(w, w') = \lvert N_H(w) \triangle N_H(w') \rvert$ and the triangle inequality	`GsimOT.rho` Common/Lemma35_Integrality.lean:25 `GsimOT.rho_triangle` Common/Lemma35_Integrality.lean:36
Lemma 29 (first part)	28	$Q(v, w) > 0 \Rightarrow \rho_H(\sigma(v), w) \leq 2L$	`GsimOT.rho_le_localCost_add` Common/Lemma35_Integrality.lean:43 `GsimOT.lemma35_rho_le` Common/Lemma35_Integrality.lean:110
Lemma 29 (second part)	28	$L < 1 \Rightarrow \sigma$ is an isomorphism	`GsimOT.rho_eq_zero_of_deg_eq_of_lt_two` Common/Lemma35_Integrality.lean:79 `GsimOT.lemma35_iso` Common/Lemma35_Integrality.lean:120 `GsimOT.lemma35_dOne_eq_zero` Common/Lemma35_Integrality.lean:162
Lemma 29 ("in particular")	28	$G \ncong H \Rightarrow d_1^{\mathrm{OT}}(G, H) \geq 1$	`GsimOT.lemma35_one_le_dOTOne` Common/Lemma35_Integrality.lean:176

Operator norms (`OperatorNorm.lean`, `Attainment.lean`)

Paper	Page	Statement	Lean declaration(s) and location

App. A	14	ℓ^p norm $\|x\|_p$ $(1 \le p \le \infty)$	`GsimOT.lpNorm`
			`Common/OperatorNorm.lean:25`
			`GsimOT.lpNorm_eq_sum`
			`Common/OperatorNorm.lean:57`
			`GsimOT.lpNorm_top`
			`Common/OperatorNorm.lean:62`

App. A	14	Operator norm $\|A\|_{p\to q} = \max_{\|x\|_p=1} \|Ax\|_q$ (in Lean the supremum over the unit ball $\|x\|_p \le 1$, the same value for nonempty index sets)	`GsimOT.opNorm`
			`Common/OperatorNorm.lean:28`
			`GsimOT.lpNorm_mulVec_le`
			`Common/OperatorNorm.lean:128`
			`GsimOT.opNorm_le_of_bound`
			`Common/OperatorNorm.lean:135`
			`GsimOT.exists_lpNorm_mulVec_eq_opNorm`
			`Common/OperatorNorm.lean:142`

App. A	14	Hölder conjugate q^* with $1/q^* = 1 - 1/q$	`GsimOT.one_le_conjExponent`
			`Common/OperatorNorm.lean:33`
			`GsimOT.inv_conjExponent_toReal`
			`Common/OperatorNorm.lean:38`

App. A	14	Hölder's inequality $\|y^\top z\| \le \|y\|_{q^*}\|z\|_q$ and its equality case	`GsimOT.abs_dotProduct_le_lpNorm_mul`
			`Common/OperatorNorm.lean:188`
			`GsimOT.exists_dotProduct_eq_lpNorm`
			`Common/OperatorNorm.lean:224`

Lemma 19	14	$\|A\|_{p\to q} = \max_{\|y\|_{q^*}=1,\|x\|_p=1} \|y^\top Ax\|$ (in Lean over the unit balls)	`GsimOT.lemma24`
			`Common/OperatorNorm.lean:290`
			`GsimOT.abs_dotProduct_mulVec_le_opNorm`
			`Common/OperatorNorm.lean:267`
			`GsimOT.exists_abs_dotProduct_mulVec_eq_opNorm`
			`Common/OperatorNorm.lean:281`

Paper	Page	Statement	Lean declaration(s) and location

–	–	Triangle inequality, $\|A\|_{p\to q} \le \sum_{i,j} \|A_{ij}\|$ and continuity of $\|\cdot\|_{p\to q}$	`GsimOT.opNorm_add_le`
			`Common/OperatorNorm.lean:169`
			`GsimOT.opNorm_le_sum_abs`
			`Common/OperatorNorm.lean:158`
			`GsimOT.abs_opNorm_sub_le`
			`Common/Attainment.lean:78`
			`GsimOT.continuous_opNorm`
			`Common/Attainment.lean:91`

App. A	15	Restricting to rows and columns "cannot increase the operator norm"	`GsimOT.lpNorm_le_of_abs_le`
			`Common/Attainment.lean:39`
			`GsimOT.opNorm_restrict_le`
			`Common/Attainment.lean:53`

Cut norm and cut distance (`CutNorm.lean`, `AlignedDistances.lean`)

Paper	Page	Statement	Lean declaration(s) and location

App. A	14	Cut norm $\|A\|_\square = \max_{S,T} \|A(S,T)\|$	`GsimOT.cutNorm`
			`Common/CutNorm.lean:32`
			`GsimOT.le_cutNorm`
			`Common/CutNorm.lean:39`
			`GsimOT.cutNorm_le`
			`Common/CutNorm.lean:44`
			`GsimOT.exists_eq_cutNorm`
			`Common/CutNorm.lean:52`
			`GsimOT.cutNorm_smul`
			`Common/CutNorm.lean:59`

Sec. 1, Sec. 2	2, 5	$B(G,H,Q)$ and $C_\square(Q) = \|B(G,H,Q)\|_\square$	`GsimOT.B`
			`Common/OTDistances.lean:58`
			`GsimOT.costCut`
			`Common/CutNorm.lean:79`

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, Sec. 2	1–2, 5	$d_\square$, $d_\square^{\mathrm{OT}}$ (infimum) and $\delta_\square = n^{-2} d_\square$, $\delta_\square^{\mathrm{OT}} = n^{-2} d_\square^{\mathrm{OT}}$	GsimOT.dCut Common/CutNorm.lean:82 GsimOT.dOTCut Common/CutNorm.lean:85 GsimOT.deltaCut Common/CutNorm.lean:88 GsimOT.deltaOTCut Common/CutNorm.lean:91
App. E.2, Eq. (21), E.2.2	32, 36	$\zeta_R(A,B) = \|D_{A,B}\|_{\square,\mu_R}$ with $\mu_R = R/n$; $\zeta_R(A_G, A_H) = \zeta_R = n^{-2} C_\square(R)$ for graphs	GsimOT.epsR Common/CutNorm.lean:105 GsimOT.adjMat Common/CutNorm.lean:94 GsimOT.epsR_adjMat Common/CutNorm.lean:110
Sec. 1, App. B	1, 3, 15	$\kappa_{p\to q} = 1/q + 1 - 1/p$, $A_G^\pi - A_H$, $d_{p\to q}$ and $\delta_{p\to q} = n^{-\kappa_{p\to q}} d_{p\to q}$ $(s(n) = n^{\kappa_{p\to q}})$	GsimOT.kappa Common/AlignedDistances.lean:23 GsimOT.alignedDiff Common/AlignedDistances.lean:31 GsimOT.distPQ Common/AlignedDistances.lean:35 GsimOT.deltaPQ Common/AlignedDistances.lean:39

Optimal transport distances (`OTDistances.lean`, `Attainment.lean`)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, App. A	2–3, 14	Couplings $\mathrm{DS}^*(V,W)$; x^a with $0^a := 0$ (convention $0^0 = 0$)	GsimOT.IsCoupling Common/OTDistances.lean:36 GsimOT.powZ Common/OTDistances.lean:42

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, Eqs. (1), (2)	2–3	$B(G,H,Q)$ and $B_{p\to q}(G,H,Q)$	GsimOT.B Common/OTDistances.lean:58 GsimOT.Bmat Common/OTDistances.lean:70
Eq. (2), Sec. 2	3, 5	$\delta^{\mathrm{OT}}_{p\to q}$ (couplings) and $d^{\mathrm{OT}}_{p\to q}$ ($\mathrm{DS}(V,W)$), both infima	GsimOT.deltaOTPQ Common/OTDistances.lean:74 GsimOT.dOT Common/OTDistances.lean:79
Secs. 1–2, Apps. A–B	3, 5, 14–15	$\delta^{\mathrm{OT}}_{p\to q} = n^{-\kappa_{p\to q}} d^{\mathrm{OT}}_{p\to q}$ for graphs of the same order n ($d^{\mathrm{OT}}_{p\to q}$ is the unnormalised version, $s(n) = n^{\kappa_{p\to q}}$), via $\mathrm{DS}(V,W) = n\,\mathrm{DS}^*(V,W)$ and the homogeneity of $B_{p\to q}$ and of $\|\cdot\|_{p\to q}$	GsimOT.deltaOTPQ_eq_dOT Common/OTDistances.lean:131 GsimOT.Bmat_smul Common/OTDistances.lean:117 GsimOT.opNorm_smul_of_nonneg Common/OTDistances.lean:94
App. A	14–15	"All minima … are attained" for $\delta^{\mathrm{OT}}_{p\to q}$ and $d^{\mathrm{OT}}_{p\to q}$: lower semicontinuity of $Q \mapsto \|B_{p\to q}(G,H,Q)\|$, compactness	GsimOT.lowerSemicontinuous_opNorm_Bmat Common/Attainment.lean:136 GsimOT.isCompact_setOf_isCoupling Common/Attainment.lean:168 GsimOT.exists_deltaOTPQ_eq Common/Attainment.lean:199 GsimOT.exists_dOT_eq Common/Attainment.lean:223

Lemma 22 for the edit distance (`Lemma27_LockingMarkersEdit.lean`)

Paper	Page	Statement	Lean declaration(s) and location
Lemma 22 (1), $\diamondsuit = \mathrm{ed}$	21	$C_{\mathrm{ed}}(\pi) < \tau \Rightarrow \pi$ fixes all markers and $\pi(V) = V$; via $C_1(\pi) \le C_{\mathrm{ed}}(\pi)$ and the case $\diamondsuit = 1$	GsimOT.costOneBij_le_costEditBij Common/Lemma27_LockingMarkersEdit.lean:26 GsimOT.lemma27_bij_edit Common/Lemma27_LockingMarkersEdit.lean:41

Lemma 22 (2), $\diamondsuit = \mathrm{ed}$	21	$Q(z_j, z_j) = 1 \Rightarrow C_{\mathrm{ed}}(Q')$ w.r.t. (G', H') equals $C_{\mathrm{ed}}(Q)$ w.r.t. (G, H)	GsimOT.lemma27_cost_edit Common/Lemma27_LockingMarkersEdit.lean:53

Edit distance and Eq. (3) (EditDistance.lean, BirkhoffDecomposition.lean, MatrixNorms.lean)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, Sec. 2	2, 5	$C_{\mathrm{ed}}(Q) = \sum_{v,w,v',w'} \lvert D(v,w,v',w')\rvert\, Q(v,w)\, Q(v',w')$ and $C_{\mathrm{ed}}(\pi) = C_{\mathrm{ed}}(P_\pi) = \lVert A_G^\pi - A_H \rVert_{\mathrm{ed}}$	GsimOT.costEdit Common/EditDistance.lean:35 GsimOT.costEditBij Common/EditDistance.lean:39 GsimOT.costEditBij_eq Common/EditDistance.lean:60
Sec. 1, Eq. (1), Sec. 2, App. A	2, 5, 14–15	$d_{\mathrm{ed}} = \min_\pi C_{\mathrm{ed}}(\pi)$ and $d_{\mathrm{ed}}^{\mathrm{OT}} = \min_{Q \in \mathrm{DS}(V,W)} C_{\mathrm{ed}}(Q)$ (infimum, attained); $d_{\mathrm{ed}}^{\mathrm{OT}} \le d_{\mathrm{ed}}$	GsimOT.dEdit Common/EditDistance.lean:42 GsimOT.dOTEdit Common/EditDistance.lean:45 GsimOT.exists_dOTEdit_eq Common/EditDistance.lean:109 GsimOT.dOTEdit_le_dEdit Common/EditDistance.lean:102
Sec. 1, Sec. 2	1–2, 5	$\delta_{\mathrm{ed}} = n^{-2} d_{\mathrm{ed}}$, $\delta_{\mathrm{ed}}^{\mathrm{OT}} = n^{-2} d_{\mathrm{ed}}^{\mathrm{OT}}$ (scaling by $1/n^2$)	GsimOT.deltaEdit Common/EditDistance.lean:48 GsimOT.deltaOTEdit Common/EditDistance.lean:51
Sec. 1, Sec. 2, App. A	2, 5, 14	$C_{\mathrm{ed}}(Q) = \lVert B(G,H,Q) \rVert_{\mathrm{ed}}$ for $Q \ge 0$ (entrywise 1-norm $\lVert\cdot\rVert_{\mathrm{ed}} = \lVert\cdot\rVert_{(1)}$)	GsimOT.entrywiseNorm Common/MatrixNorms.lean:53 GsimOT.costEdit_eq_entrywiseNorm_B Common/MatrixNorms.lean:119
Sec. 2, Eq. (3)	5	Birkhoff–von Neumann: $Q = \sum_i \lambda_i P_{\pi_i}$ with $\lambda_i > 0$, $\sum_i \lambda_i = 1$ (also: over all $\sigma \in \mathrm{Perm}(V)$ with $\lambda \ge 0$); conversely every such combination is in $\mathrm{DS}(V,W)$	GsimOT.exists_birkhoff_decomposition_pos Common/BirkhoffDecomposition.lean:72 GsimOT.exists_birkhoff_decomposition Common/BirkhoffDecomposition.lean:42 GsimOT.isDS_of_decomposition Common/BirkhoffDecomposition.lean:23

Eq. (7) (`MatrixNorms.lean`)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 2, Eq. (7)	5, 14	$\|B_{1\to1}(G,H,Q)\|_{1\to1} = C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$ for $Q \in \mathrm{DS}(V,W)$; hence $d^{\mathrm{OT}}_{1\to1} = d^{\mathrm{OT}}_1$	`GsimOT.opNorm_Bpq_one_one_eq_costOne` `Common/MatrixNorms.lean:150` `GsimOT.dOT_one_one_eq_dOTOne` `Common/MatrixNorms.lean:164`

Doubly stochastic matrices (`DoublyStochastic.lean`)

Paper	Page	Statement	Lean declaration(s) and location
–	–	Sums over $V \times W$ against $Q \in \mathrm{DS}(V,W)$; partial sums $Q(S,T)$ and their bounds; $\mathrm{DS}(V,W)$ is compact; bounds for infima over $\mathrm{DS}(V,W)$	`GsimOT.IsDS.sum_mul_fst` `Common/DoublyStochastic.lean:35` `GsimOT.mass` `Common/DoublyStochastic.lean:118` `GsimOT.card_mul_le_of_lt_mass` `Common/DoublyStochastic.lean:150` `GsimOT.isCompact_setOf_isDS` `Common/DoublyStochastic.lean:77` `GsimOT.iInf_isDS_le` `Common/DoublyStochastic.lean:98` `GsimOT.le_iInf_isDS` `Common/DoublyStochastic.lean:104`

Graphs of the reductions of Section 4 (`PathsAndCycles.lean`, `ThreePartition.lean`, `Leaves.lean`)

Paper	Page	Statement	Lean declaration(s) and location
Proofs of Thms. 15–17	50–57	Disjoint unions of paths (F_1) and of cycles; their degrees and edge counts	`GsimOT.Hardness.pathUnion` Common/PathsAndCycles.lean:56 `GsimOT.Hardness.cycleUnion` Common/PathsAndCycles.lean:226 `GsimOT.Hardness.sum_adj_pathUnion` Common/PathsAndCycles.lean:134 `GsimOT.Hardness.deg_cycleUnion` Common/PathsAndCycles.lean:238
Proof of Thm. 15	50	Three-Partition instances $(A, a_1, \dots, a_{3m})$; the step of adding 3 to each a_i and 9 to A	`GsimOT.Hardness.ThreePartition` Common/ThreePartition.lean:26 `GsimOT.Hardness.ThreePartition.IsYes` Common/ThreePartition.lean:39 `GsimOT.Hardness.ThreePartition.shift` Common/ThreePartition.lean:206 `GsimOT.Hardness.ThreePartition.shift_isYes_iff` Common/ThreePartition.lean:223
Proofs of Thms. 15–17	50–56	A solution gives a bijection mapping each path into one component; conversely a $Q \in \mathrm{DS}$ mapping every path into a single component gives a solution (counting mass)	`GsimOT.Hardness.ThreePartition.layoutEquiv` Common/ThreePartition.lean:137 `GsimOT.Hardness.ThreePartition.isYes_of_isDS` Common/ThreePartition.lean:168
Proofs of Thms. 17, 18	54, 57	Attaching $\ell(b)$ leaves to every vertex b of a base graph	`GsimOT.Hardness.addLeaves` Common/Leaves.lean:33

Notes on the formalisation

1. The paper defines $C_1(Q)$ as the operator $1 \to 1$-norm of $B_{1\to1}(G, H, Q)$ and gives the formula $C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v, w)$ (Sec. 2, p. 5, l. 184–189; Eq. (7), p. 14). The formalisation takes this formula as the definition of `costOne` and proves the identity (`opNorm_Bpq_one_one_eq_costOne`).

2. d_1^{OT}, $d_{\mathrm{ed}}^{\mathrm{OT}}$, $d_{\square}^{\mathrm{OT}}$, $d_{p\to q}^{\mathrm{OT}}$ and $\delta_{p\to q}^{\mathrm{OT}}$ are defined as *infima*. The paper takes minima and argues that they are attained (App. A, pp. 14–15, l. 503–510). This is formalised for the operator norms (`exists_deltaOTPQ_eq`, `exists_dOT_eq`, following the paper's lower-semicontinuity argument; used by Claim 20) and for the edit distance (`exists_dOTEdit_eq`, used by Theorem 15). For bijections the minimum over the finite set $\mathrm{Bij}(V, W)$ is the infimum.

3. Normalisation: for graphs of the same order n, $\delta_{\diamond} = d_{\diamond}/s(n)$ and $\delta_{\diamond}^{\mathrm{OT}} = d_{\diamond}^{\mathrm{OT}}/s(n)$ with $s(n) = n^2$ for the edit and the cut distance (p. 1, l. 24–27) and $s(n) = n^{\kappa_{p\to q}}$, $\kappa_{p\to q} = 1/q + 1 - 1/p$, for $\|\cdot\|_{p\to q}$ (p. 3, l. 88–90; App. B, p. 15, l. 513–514), so $s(n) = n$ for $\|\cdot\|_1$. In Lean, `deltaPQ` multiplies $d_{p\to q}$ by $n^{-\kappa}$, the other normalised distances divide by n or n^2, and `deltaOTPQ_eq_dOT` proves $\delta_{p\to q}^{\mathrm{OT}} = n^{-\kappa} d_{p\to q}^{\mathrm{OT}}$.

4. Lemma 22 is stated in the paper for $\diamondsuit \in \{\mathrm{ed}, 1\}$; both cases are formalised, $\diamondsuit = 1$ in `Lemma27_LockingMarkers.lean` and $\diamondsuit = \mathrm{ed}$ in `Lemma27_LockingMarkersEdit.lean`. In Lean, the clique vertex y_{i+1} is `Sum.inr i` and the marker index j of the paper is `j.val + 1`.

5. Lemma 29 is proved in the paper for an optimal Q and $L = d_1^{\mathrm{OT}}(G, H)$. Its proof only uses that all supported local costs of Q are at most L, which is how it is stated in Lean. It is shared by Proposition 8 (via Lemma 27) and Theorem 9.

6. Eq. (3): `exists_birkhoff_decomposition` sums over all permutations with weights ≥ 0; `exists_birkhoff_decomposition_pos` is the exact statement with $\lambda_i > 0$.